

Current Biology

Wolves use diverse tactics to track partially migratory prey

Highlights

- Wolves used diverse movement tactics to track partially migratory elk
- Migratory packs with pups shifted homesites toward migratory elk summer range
- Migratory ungulates may not be able to escape predators exhibiting migratory coupling
- Shifting ungulate migrations may impact predator movements and ecosystem dynamics

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In brief

Shawler et al. highlight dynamic predator responses to prey with variable migratory behavior. Wolves tracked partially migratory elk using diverse movement tactics, including migratory coupling. Some wolves moved young pups to homesites closer to migratory elk summer range, challenging assumptions about escaping predation as a benefit of ungulate migration.



Report

Wolves use diverse tactics to track partially migratory prey

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SUMMARY

Some predators move long distances to track migratory prey.^{1–3} This phenomenon, known as migratory coupling,¹ has been observed in wolves that track migratory caribou in the Arctic.^{4–7} By contrast, most wolves elsewhere are highly territorial and thought to be non-migratory—particularly during the denning and pup-rearing season.^{8,9} Recent advances in wildlife tracking have illuminated widespread individual variation in seasonal movements of partially migratory prey,^{10–18} but little is known about how wolves respond to such varied behaviors. Using concurrent GPS data (2019–2021) from 19 gray wolves (*Canis lupus*) and 99 elk (*Cervus canadensis*) from a large, partially migratory herd in the eastern Greater Yellowstone Ecosystem, we quantified wolf movement strategies based on overlaps between wolf packs' seasonal ranges and between wolf and elk summer ranges. We found that wolves exhibited a range of movement responses, including migrating, commuting, and remaining resident. Additionally, some packs moved pups from den sites to rendezvous sites along elk migration routes, or even within elk summer range, challenging the long-accepted notion that juvenile carnivores act as spatial anchors. The diverse strategies wolves employ to track migratory prey highlight how the plasticity and variation of migratory behaviors in their ungulate prey can translate to dynamic wolf space use. This suggests that changes in ungulate mobility—for instance, due to climate or land-use change—may reverberate across trophic levels, reshaping predator movement patterns and influencing broader ecosystem dynamics.

RESULTS

Predators across many taxa and ecosystems employ diverse strategies to exploit migratory prey.¹ Some adopt large-scale movement strategies to track seasonal resource pulses from migratory prey, a phenomenon known as migratory coupling.¹ Red knots (*Calidris canutus rufa*) will detour 1,000 km to feed on horseshoe crab (*Limulus polyphemus*) eggs,² and salmon sharks (*Lamna ditropis*) travel thousands of kilometers to reach spawning Pacific salmon (*Oncorhynchus* spp.).³ Other predators move with prey from one seasonal range to another, such as the gray wolves (*Canis lupus*) that follow migrating barren-ground caribou (*Rangifer tarandus groenlandicus*) across the Canadian Arctic tundra.^{4–7} However, most wolves in temperate regions are considered highly territorial and non-migratory, especially during denning and pup-rearing.^{8,9} These contrasting behaviors highlight substantial variation in movement strategies within a single predator species, yet the reasons for such divergent behaviors remain poorly understood.

This variation in predator behavior is further complicated by the fact that prey migrations themselves are often highly variable and unpredictable. Growing evidence shows that many migratory ungulate populations are partially migratory, and some individuals migrate while others remain resident.^{10–12} Moreover, recent studies have documented considerable plasticity in ungulate migration behavior—members of the same group may migrate to different places, leave at different times, or even switch between migration and residency.^{13–18} Despite increasing recognition of this complexity in prey movements, little is known about how predators respond to such variable and flexible migratory strategies.

The Greater Yellowstone Ecosystem (GYE) provides a prime opportunity to examine predator response to variability in migratory prey behavior through wolf-elk interactions. The GYE hosts 18 herds of partially migratory elk (*Cervus canadensis*) with diverse migratory tactics that vary in propensity, routes, timing, and distance.^{17–21} Wolves in the GYE rely on elk as their primary prey, and many packs establish territories and dens on low-elevation elk winter range outside of Yellowstone National



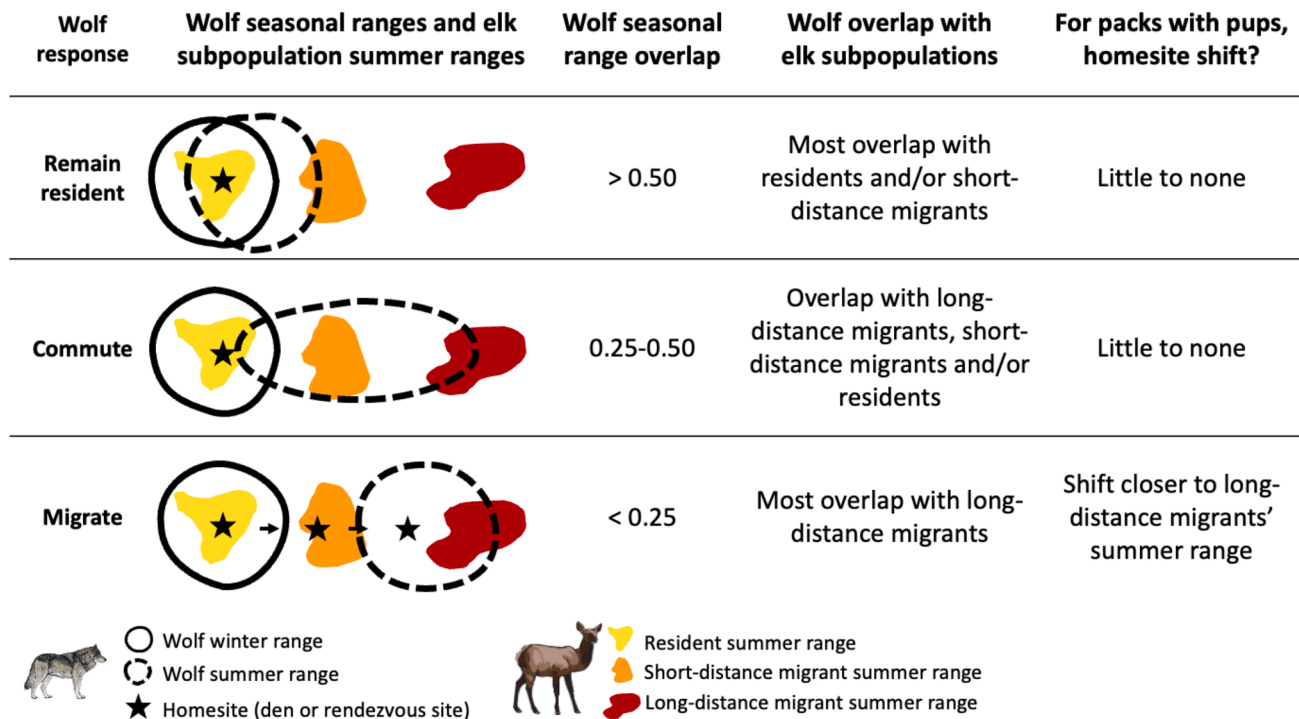


Figure 1. Summary of wolf response predictions and analytical criteria used to assess whether wolves remain resident, commute, or migrate in response to elk movement strategies

Park. Most of these wolves are assumed to be non-migratory and constrained to pups at homesites when elk are migrating during spring.²² One previous study found that migrating and resident elk within the same herd elicited different wolf movement responses,²³ and advances in tracking technology now present an opportunity to examine predator responses to the full portfolio of ungulate movement tactics.

Our investigation uncovers surprisingly diverse and dynamic movement strategies adopted by wolves in response to the seasonal movements of a partially migratory elk herd with a wide range of individual behaviors. Within this elk herd, we identified five elk subpopulations, classified by their migratory behavior and geographic distribution: long-distance north, long-distance south (i.e., long-distance migratory elk; 80–180 km seasonal movements); short-distance north, short-distance south (i.e., short-distance migratory elk; 30–60 km seasonal movements); and resident (i.e., elk that did not migrate). Based on seasonal wolf distributions and overlap between wolf and elk summer ranges, we hypothesized wolves would adopt one of three strategies to track the different elk subpopulations: (1) remain resident, (2) commute to and from elk summer ranges, or (3) migrate.^{5,7,23–25} Additionally, to evaluate the degree to which wolf pups act as a spatial constraint decoupling wolf packs from migratory elk, we examined whether wolves shifted homesite locations (i.e., dens and rendezvous sites, the latter being areas where packs relocate once pups are old enough to leave the den and serve as gathering points for the pack). We documented shifts of homesites by mapping their locations, recording shift dates, calculating the distance of each shift, and noting whether the shift was toward an elk subpopulation summer

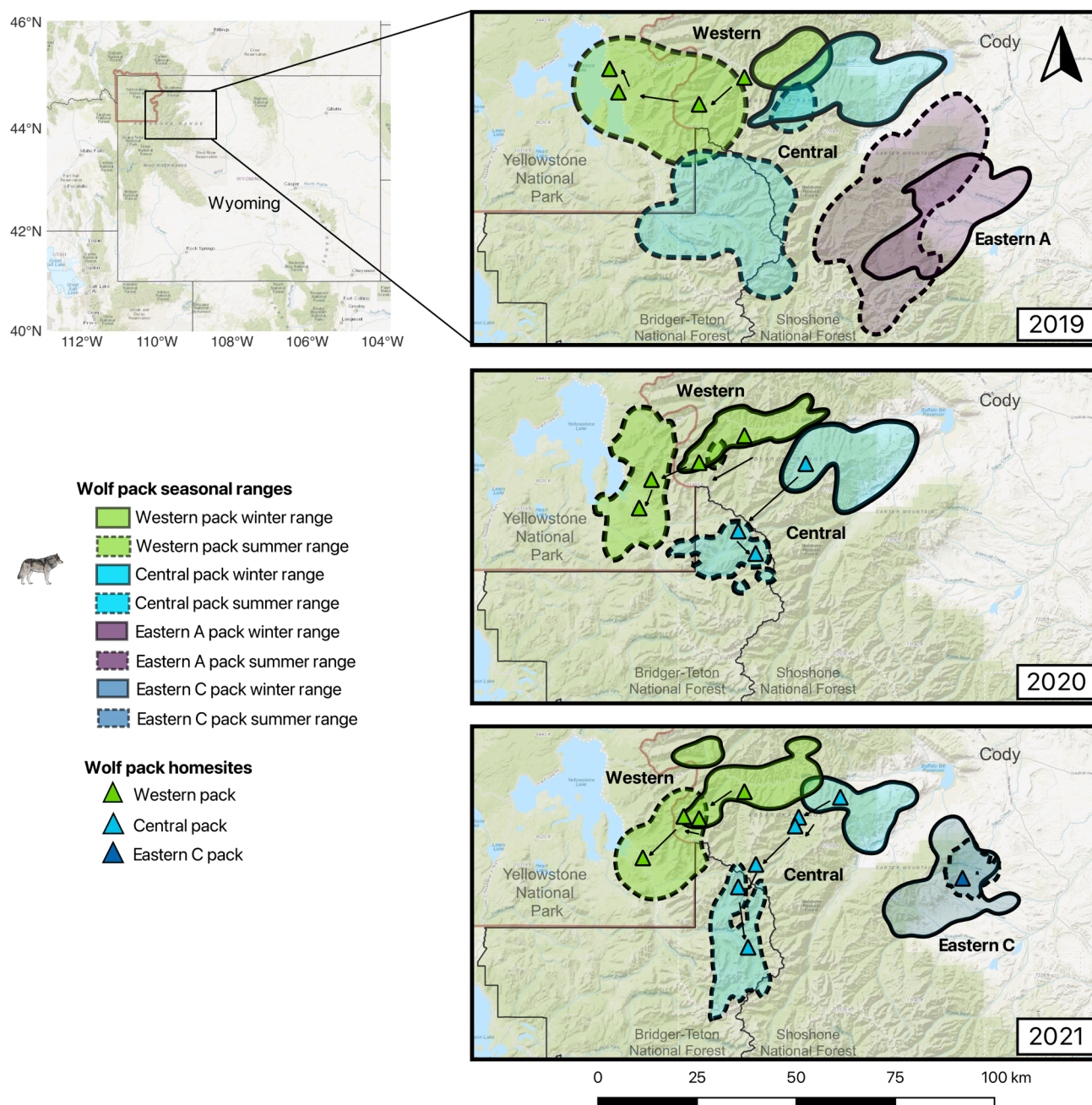
range. Predictions (Figure 1) were tested using GPS collar locations from sympatric wolves ($n = 19$) and elk ($n = 99$) recorded over 3 years in the eastern GYE.

Wolves exhibited different types of seasonal movements

Overlap of wolf winter (February 14 to March 14) and summer (August) ranges delineated by autocorrelated kernel density estimation (AKDE) shows the extent to which wolves shifted their range distribution from winter to summer (Figures 2 and S1; Table S3). Overlap values in our study area varied considerably among packs. Of the four packs we studied, two had little to no overlap between their summer and winter ranges for all 3 years of the study, which supports migratory behavior according to our predictions in Figure 1. The Central pack had 0.12 overlap (95% CI 0.1, 0.13) in 2019 and 0 overlap in other years, and the Western pack had 0.04 overlap (95% CI 0, 0.3) in 2019, 0.06 overlap (95% CI 0.05, 0.07) in 2020, and 0.08 overlap (95% CI 0.03, 0.16) in 2021. Conversely, two packs had high (0.54 [95% CI 0.51, 0.56], Eastern C pack in 2021) to moderate (0.44 [95% CI 0.42, 0.46], Eastern A pack in 2019) levels of overlap. Eastern A pack in 2020 and Eastern B pack in 2021 had insufficient data to perform seasonal overlap analyses due to collared individuals either dispersing out of the study area, collar failures, or lethal removals prior to the summer season.

Wolves tracked different elk subpopulations in the summer

Both the Western and Central packs mainly overlapped with long-distance migratory elk, while the Eastern packs mainly overlapped



with short-distance migratory and resident elk ([Figures 3](#) and [S2](#); [Table S4](#)), which supports migratory and resident behavior according to our predictions in [Figure 1](#). The Western pack overlapped the most with long-distance north elk and had moderate overlap with long-distance south elk. The Central pack mostly overlapped with long-distance south elk, with some overlap with long-distance north and resident elk. Eastern A pack in 2020 and Eastern B pack in 2021 did not have sufficient data to perform

overlap analyses, but Eastern A pack in 2019 mostly overlapped with short-distance south and resident elk, while Eastern C pack in 2021 overlapped the most with resident elk.

Wolves shifted homesites to track long-distance migratory elk

In contrast to the common assumption that wolves are spatially anchored to homesites during the pup-rearing season, we

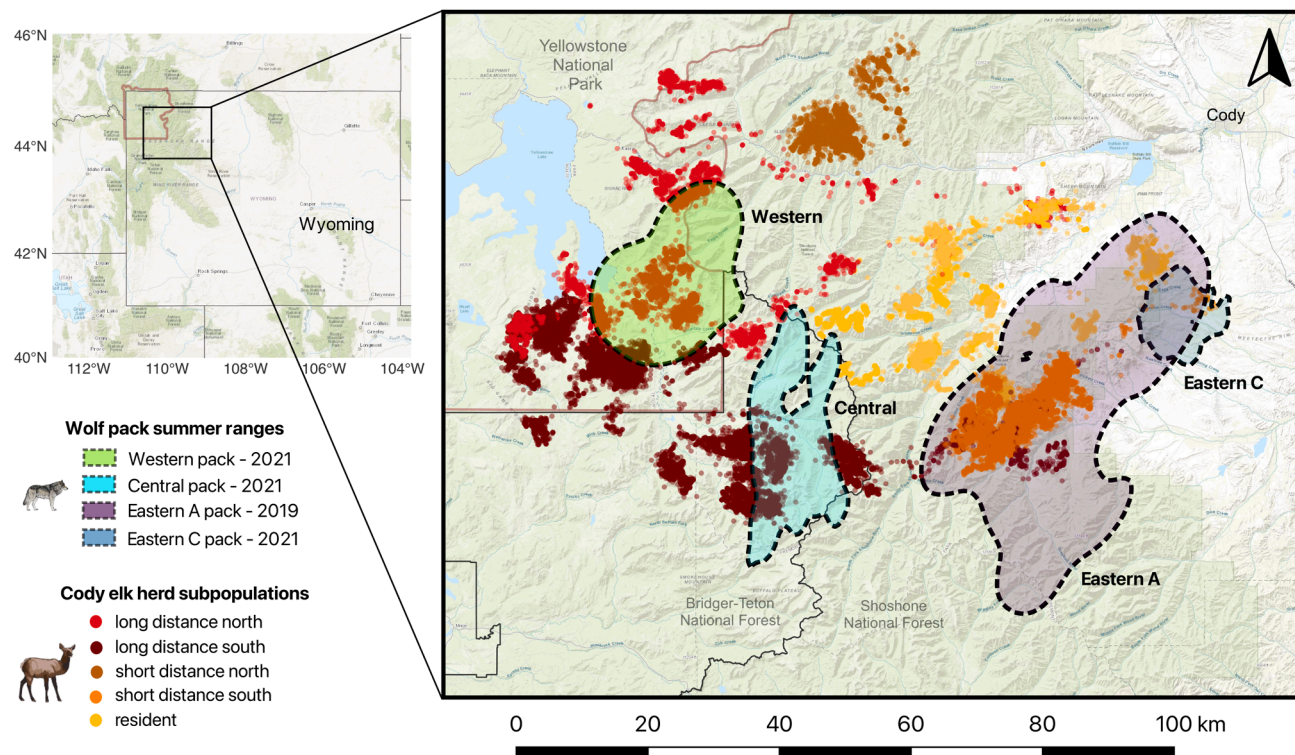


Figure 3. Overlap of wolf packs and elk subpopulations during the summer range (August)

Wolf pack summer ranges are derived from 95% AKDEs and elk subpopulations are represented by GPS points. Western pack, Central pack, and Eastern C pack summer ranges are from 2021, while Eastern A pack is from 2019. Elk summer range locations are from 2019, 2020, and 2021. See also [Figure S2](#) and [Table S4](#).

documented wolves making long-distance homesite shifts with young pups toward long-distance migratory elk summer range ([Figures 2](#) and [4](#); [Table S5](#)), providing additional evidence supporting migratory behavior according to [Figure 1](#). The Western pack had pups in all 3 years of the study and shifted from the den site to multiple rendezvous sites in all years. The Western pack's den site and first rendezvous site (R1) are 13.2 km apart (Euclidean distance) and were used in all years of the study. The Western pack's final rendezvous sites for 2020 and 2021 were within the long-distance north elk summer range (due to collar failure in 2019, the final rendezvous site was unknown). The Central pack had pups in 2020 and 2021 and shifted from the den site to multiple rendezvous sites closer to the long-distance south elk summer range. The longest shift (23.92 km) was from their den site to R1 in 2020. Of the three Eastern packs, only Eastern C pack had pups. Their two rendezvous sites were very close to the den site ($R1 = 2.74$ km; $R2 = 0.69$ km), and all homesites were within resident elk summer range and near short-distance south elk summer range. However, they never made complete shifts and often returned to the den site with pups.

DISCUSSION

Investigating how predators respond to prey with variable and flexible movement strategies is a vital first step to understanding how shifting predator-prey interactions influence trophic

structure and broader ecosystem dynamics. We evaluated how wolves changed their space use in response to seasonal changes in elk availability where they had access to a large herd that exhibited a wide range of migratory tactics. Wolves in our study area employed a variety of behavioral responses to the seasonal movements of a partially migratory elk herd, from remaining resident to migrating long distances into elk summer ranges. However, contrary to our predictions in [Figure 1](#) that each wolf pack would employ a single behavioral strategy, we found that some packs exhibited combinations of behaviors. In two instances, packs met two of the three criteria in [Figure 1](#) for migratory and resident behavior, respectively, while also exhibiting one criterion associated with commuting. Wolves that exhibited migratory coupling moved between distinct, and in some cases completely separate, winter and summer ranges ([Figure 2](#)). Our post hoc analysis of overlap between the winter range and each monthly range from March to December revealed wolves began decreasing overlap in late spring and early summer, which coincides with when wolves embarked on commutes before fully migrating later in the summer ([Figure S2](#)). Therefore, commuting may be a crucial behavior enabling full migratory movements by acting as scouting forays for wolves to track migratory elk more effectively. Since elk can vary migration strategies,^{17,18} these costly forays may help wolves align the timing of long-distance shifts with elk availability. Wolves could also be using commutes to track elk environmental cues (e.g., greenwave or snowmelt), which may prove more predictable

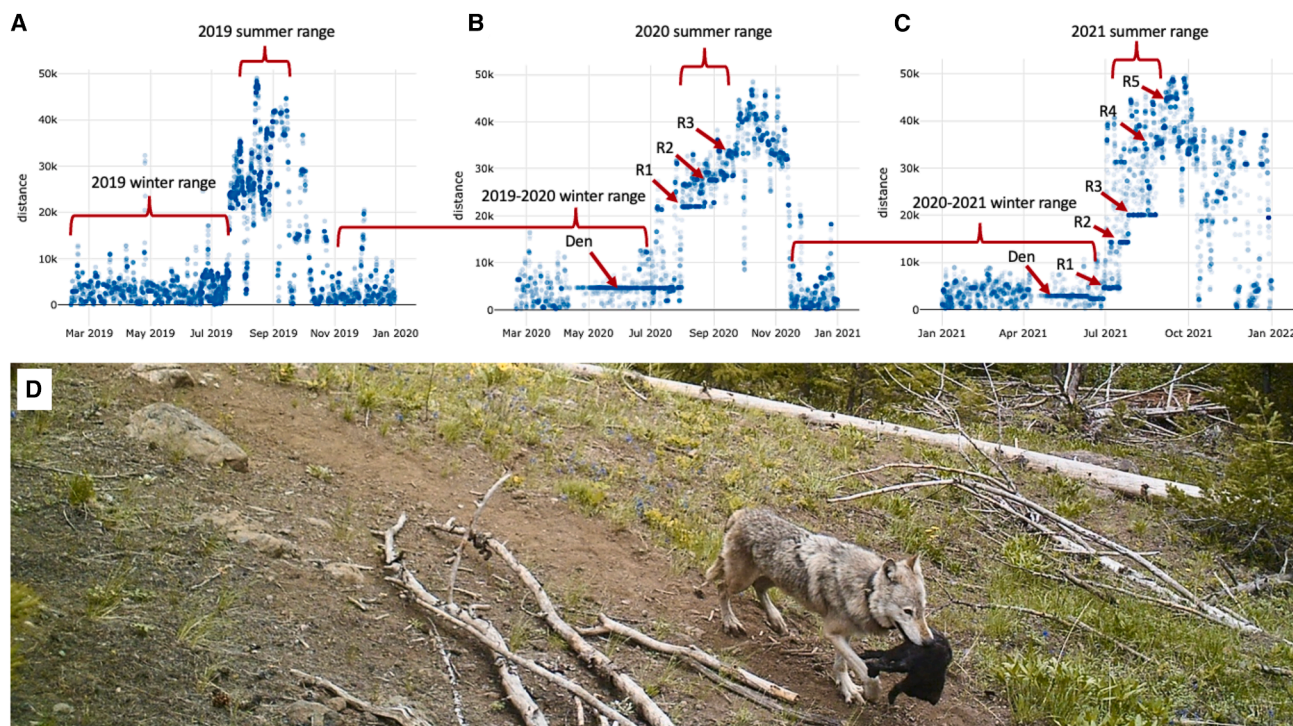


Figure 4. Visual depictions of the seasonal movements of wolves between winter range, summer range, and homesites (den and rendezvous sites)

(A–C) The top plots show net-square displacement curves of the Central pack's seasonal movements in 2019 (A) when the pack did not have pups and in 2020 (B) and 2021 (C) when the pack did have pups.

(D) Trail camera photo of the Western pack breeding female carrying a young pup from the den site to the rendezvous site in early June 2020. The pup is estimated to be between 1 and 2 months old in this photo (photo provided by Wyoming Game and Fish Department).

than elk themselves.¹⁷ Thus, while the increasing unpredictability of prey migratory behavior may threaten to decouple predators from their prey, wolves may use commuting as a way to cope with this unpredictability.

Our results highlight that even within the same landscape, predators can differ in their movement strategies—raising the question of why some groups engage in migratory coupling while others do not. Three factors may explain this variation. First, migratory coupling may be associated with predator stability and longevity. The three wolf packs in our study area that remained resident experienced high rates of turnover due to human-caused mortality (e.g., hunting and lethal control) and were likely less familiar with the landscape and elk movement behaviors. Second, high predator population density may preclude the ability to track migratory prey across long distances. For example, wolves in Yellowstone National Park adjacent to our study area occur at very high densities, requiring them to maintain and defend consistent annual territories against neighboring packs rather than moving seasonally across the landscape.^{9,26} Future studies, especially in regions with a gradient of predator densities, could further explore how population density shapes a predator's tendency to engage in migratory coupling. We observed resident wolves commuting into migratory elk summer range but not fully migrating, suggesting the presence of migratory wolves in these areas may have deterred residents from moving into elk summer range. Third, alternate

prey availability could decrease the propensity of predators to engage in migratory coupling (i.e., prey switching²⁷). Resident wolves in our study may not follow migratory prey because they have access to resident elk and other prey species that don't embark on long-distance migrations. These wolves had the greatest overlap with resident and short-distance migrant elk. Our related wolf predation and diet study found a greater percentage of biomass of alternate prey (e.g., deer [*Odocoileus sp.*], pronghorn [*Antilocapra americana*], and cattle [*Bos taurus*]) for resident wolves compared with migratory wolves.²⁸

Our findings challenge the long-held assumption that elk and other ungulates can effectively minimize predation risk by migrating to high-elevation areas, spatially separating from wolves that are anchored to immobile pups throughout the pup-rearing period.²² On the contrary, we found instances of packs with pups tracking elk on long-distance migrations and packs without pups remaining resident. While wolves tracking migratory prey with pups have been observed in the Arctic regions of North America, these movements occurred in late summer when pups were more mobile.^{4,7,29} Additionally, wolves in the continental U.S. that have been observed shifting rendezvous sites 1–8 km did so later in the summer.^{8,30} To our knowledge, this study is the first to show wolves with young pups shifting rendezvous sites closer to migratory prey in spring and early summer. In every year of our study, the Western pack moved approximately 1-month-old pups over 13 km

to the same rendezvous site—a pattern they had been following for several years before our observations began, according to local Wyoming Game and Fish Department wolf biologists (Figure 4). Consistently using the same homesites may make long-distance homesite shifts less risky and worth the benefit of being closer to migratory elk on their summer ranges.

The predictability of movement and parturition patterns of ungulate prey may determine whether wolves make large-scale shifts before or after pups are born. In our study system, migratory elk exhibit highly variable timing from year to year (Table S1),¹⁷ which may explain why wolves wait to shift homesites until after pups are born. We found that packs with pups generally had smaller summer range sizes than packs without pups (Figure 2)—suggesting that pups do act as a spatial anchor to some degree during the summer. However, wolves found ways to cope by shifting rendezvous sites progressively closer to migratory elk summer range, reducing the need to travel long distances to search for prey and allowing wolves to concentrate movements in a smaller area. By contrast, wolves in northern Quebec made long-distance movements *before* caribou migrated and established dens near migration routes and summer ranges, enabling access to prey despite being spatially constrained by pups.⁷ This difference may be driven by the greater predictability of caribou migration and calving, which are characterized by high site fidelity, temporal consistency, and a brief calving pulse.³¹ These contrasts suggest that predator strategies may be shaped by how reliably prey movements and parturition events can be anticipated. Evaluating how both prey and predators adjust to changing resources is an aspect of migratory coupling that will prove increasingly salient in the face of ongoing environmental changes.

Where the dynamic nature of ungulate migrations gives rise to similarly flexible behavioral responses in predators, these responses may scale up to influence broader-scale ecosystem dynamics. Globally, migratory ungulates exhibit remarkable behavioral plasticity in response to changing conditions—adjusting timing, routes, and even switching between migratory and resident tactics.^{15–18,32–35} Yet despite this flexibility, ungulate migrations are declining across multiple taxa and geographic areas.^{15,16,36,37} Such losses risk disrupting predator-prey interactions and could reverberate across trophic levels, reshaping ecosystem structure and function.³⁸ For example, migratory coupling that allows predators to track increasingly unpredictable or diminished migrations could exacerbate the vulnerability of prey populations already weakened by environmental and anthropogenic stressors. For decades, ecologists have cited seasonal escape from predators as a key benefit of ungulate migration,³⁹ but we found that migration does not always allow vulnerable prey populations to escape predation risk during the pup-rearing season. Conversely, if environmental changes lead to migratory decoupling, decreased prey access for predators could reduce predator fitness, alter their distribution, and increase potential conflict with humans. Understanding migratory coupling dynamics is especially urgent where recovering carnivore populations overlap with declining migratory prey populations. Advancing our understanding of migratory coupling requires evaluating changes not only in migratory prey but also in the factors that enable or hinder predators' ability to follow them. As global change continues to disrupt migrations,

anticipating shifts in predator-prey interactions will be essential for sustaining ecological function and guiding effective conservation strategies.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to the lead contact, Avery L. Shawler (avery.shawler@gmail.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All wolf and elk GPS data have been deposited in a Mendeley Data repository and are available upon request. This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

A.L.S. and A.D.M. conceptualized the project. A.L.S. analyzed and visualized the data. A.L.S. led the writing of the paper. K.J.B., W.X., and A.D.M. contributed to writing the original draft, and T.W.M. and K.J.M. provided feedback. A.L.S. and A.D.M. managed the project. A.L.S., K.J.M., and T.W.M. collected the data. A.D.M. secured funding for the project.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
wolf and elk GPS locations	This paper	Available upon request
Experimental models: Organisms/strains		
Gray wolf (<i>Canis lupus</i>)	Cody, WY USA	N/A
Elk (<i>Cervus canadensis</i>)	Cody, WY USA	N/A
Software and algorithms		
R	https://www.r-project.org/	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animal capture and collaring

In January and February 2019 and 2020, we worked with Wyoming Game and Fish Department (WGFD) and Native Range Capture Services to deploy GPS collars on 99 individual cow elk (70 in 2019 and 29 in 2020) that were fitted with Lotek GPS Litetrack Iridium-420 collars (Newmarket, Ontario, Canada) in 2019 and Vectronic Vertex Plus GPS Iridium collars (Berlin, Germany) in 2020. We collared 19 wolves (9 in 2019, 4 in 2020, and 6 in 2021) and fitted them with Lotek GPS LiteTrack Iridium-420 collars in 2019 and Telonics Gen4 GPS-Iridium collars (Model TGW-4577-4, Telonics, Mesa, Arizona, USA) in 2020 and 2021. Wildlife captures and handling were approved by WGFD (Chapter 33 permit #1205) and University of California Berkeley's Institutional Animal Care and Use Committee AUP-2018-07-11261. Elk GPS collar recorded locations every two hours, while wolf GPS collar recorded locations every two hours during the winter and every 20 minutes to 1 hour in the summer. Elk GPS collars were deployed across the winter range of the Cody herd. We deployed GPS collars in seven wolf packs that occurred within the range of the Cody elk herd. Due to GPS collar malfunctions, human-caused wolf mortalities (e.g., harvest, poaching, lethal control), and dispersals out of the study area, five of the seven packs with GPS collared wolves were analyzed in this study. Pack names and number of collars in each pack each year are as follows: Western pack (1 in 2019, 2 in 2020, 1 in 2021), Central pack (2 in 2019, 2 in 2020, 3 in 2021), Eastern A pack (2 in 2019, 2 in 2020), Eastern B pack (2 in 2021), and Eastern C pack (2 in 2021). Only one individual wolf had continuous GPS data throughout our study.

METHOD DETAILS

Study area

This study was conducted in northwestern Wyoming east of Yellowstone National Park in the Absaroka Mountains. The elevation of the study area ranges from 1700m to 3400m, with high-elevation habitats dominated by alpine meadows, subalpine forests, and talus slopes; mid-elevation habitats dominated by Douglas fir (*Pseudotsuga menziesii*), limber pine (*Pinus flexilis*) and lodgepole pine (*Pinus contorta*) in coniferous forests and aspen (*Populus tremuloides*), cottonwood (*Populus* spp.), and shrubs in deciduous forests; and low-elevation areas dominated by sagebrush (*Artemisia* spp.) steppe, grasslands and agricultural fields. A mix of U.S. Forest Service (USFS), Bureau of Land Manager (BLM), Wyoming state land, and private land were included in this study.

Study species

We focused on the wolves that predate on the Cody elk herd, whose winter ranges are along drainages near the towns of Cody and Meeteetse. The Cody elk herd is one of the largest in the GYE with at least 6,000–7,000 predominantly migratory individuals that winter on private, state, and BLM and USFS land – often on or near working ranches and public grazing allotments. The elk that migrate cover a range of distances, some as long as 117 miles, into their summer ranges in the southeast corner of Yellowstone National Park and surrounding National Forests.¹⁷ Other subpopulations of elk complete short-distance migrations and often to the stopover sites of the long-distance migrants in the Shoshone National Forest. The wolf packs we studied occupied different areas of the Cody elk herd's winter range and varied in their overlap with human-dominated areas, with the Western and Central packs occupying more remote areas than the Eastern packs.

Classifying elk subpopulations

To capture the diversity of migratory tactics within the Cody herd, we split it into five distinct subpopulations based on migratory strategy and geographic region. We determined migratory strategy by plotting distance and elevation net-squared displacement (NSD)

curves with an interactive map interface that used the *nsd* function in the *amt* package.^{18,40} Once elk were split into migrant and resident categories, we further split the migratory elk based on migration distance (long and short) and summer range location (north and south), which resulted in five subpopulations: 1) long-distance south, 2) long-distance north, 3) short-distance south, 4) short-distance north, and 5) resident. In our study, some elk embarked on short elevational migrations, but because they remained accessible to resident wolves, we classified them as resident.

Determining seasonal range dates

To determine seasonal range dates, we calculated the mean start and end dates of the spring and fall migration of long-distance migratory elk (north and south subpopulations) between 2019 and 2021 (Table S1). We used the dates of the long-distance migrants because they comprise the majority of the Cody elk herd, and their migration represents a major seasonal shift in the distribution of prey availability for wolves. We used two methods to determine start and end dates for each individual long-distance migrant. First, we used the NSD curves plotted using the interactive interface of Zuckerman et al.¹⁸ Then following the method used by Aikens et al.⁴¹ and Rickbeil et al.,¹⁷ we visually inspected the individual NSD profiles and locations on the accompanying map to determine start and end dates for spring and fall migrations. Secondly, we used a function called *segclus2d*, which performs segmentation of bivariate time-series data into homogeneous segments using Lavielle's method – typically used to identify home range shifts.⁴² Segmenting elk GPS data using *segclus2d* allowed us to identify distinct seasonal ranges. However, there were some instances where the segments representing seasonal ranges included parts of migratory routes. Therefore, using a combination of the two methods allowed us to cross-check start and end dates. Once migration dates were determined, we identified the mean ordinal dates of the spring and fall migration start and end dates for each year of the study (similar to Anton et al.⁴³). Using the mean migration start and end dates, we determined dates for four different seasons: 1) spring migration, 2) summer range, 3) fall migration, and 4) winter range for each year (Table S1).

To ensure that our analyses reflected periods of range residency with minimal long-distance movements outside of seasonal ranges (Table S2), we selected one-month periods within the core winter and summer ranges. For winter, we chose February 14 to March 14 because the midpoint of winter range use consistently fell in mid to late February across years and also because collaring efforts for wolves and elk typically were complete by mid-February. For summer, we selected August 1 to August 31, corresponding with the typical peak of summer range residency and less long-distance movements. Table S2 shows dates of long-distance movements of the different wolf packs.

QUANTIFICATION AND STATISTICAL ANALYSIS

Calculating wolf seasonal range overlap

To assess wolf spatial shifts following elk spring migration, we calculated wolf seasonal (winter and summer) ranges using the *akde* (autocorrelated kernel density estimation (AKDE)) as this method accounts for autocorrelated GPS data and irregular sampling intervals.^{44–46} To ensure the appropriateness of *akde* for home range estimation, we conducted visual verification of range-residency via variogram analysis before fitting a series of range-resident models to the data and selecting the model that best fits the data.⁴⁷ After calculating AKDE distributions, we calculated overlap between each wolf packs' winter and summer ranges with the *overlap* function in the *ctmm* package using the Bhattacharyya coefficient, a pairwise overlap metric with a formal basis in distributional similarity that calculates overlap between density functions.⁴⁸

Because calculating winter and summer range overlap alone did not show changes in wolf distribution throughout the year, we conducted a post hoc analysis evaluating the overlap between the winter range and each monthly range from March to December (Figure S1; Table S3).

Calculating wolf and elk summer range overlap

We calculated overlap of wolf and elk summer range to evaluate how wolves track elk subpopulations after the departure of long-distance migratory elk (Figure S2; Table S4). We calculated individual elk summer range AKDEs and then calculated population range estimates for each of the five subpopulations using the *pkde* (population kernel density estimation (PKDE)) function.⁴⁵ We then used the *overlap* function to calculate overlap of wolf summer AKDEs with elk subpopulation summer PKDEs.

Documenting wolf homesite shifts

We identified homesite (i.e., den and rendezvous sites) shifts using wolf GPS collar data to plot net-square displacement curves. Den and rendezvous sites were represented by a line of points at the same distance/elevation (Figure 4). Additionally, ground-truthing by Wyoming Game and Fish Department wolf monitoring efforts and by our team during the field season further confirmed homesites. We mapped each homesite location, calculated approximate homesite shift dates, calculated the Euclidean distance between homesites, and noted whether the shift was towards a specific elk subpopulation summer range (Table S5).

Wolf behavioral response predictions

We evaluated evidence for residency, commuting, and migration based on 1) relative degrees of wolf winter and summer range overlap, 2) overlap with elk subpopulation summer range, and for packs with pups, 3) presence or lack of homesite shifts. We considered

seasonal overlaps greater than 0.5 to indicate residency, 0.25–0.5 to indicate commuting, and less than 0.25 to indicate migration. We considered most overlap with long-distance migratory elk to indicate migration, most overlap with resident/short-distance migratory elk to indicate residency and overlap with both long-distance migratory and resident/short-distance migratory elk to indicate commuting. If wolves had pups, we considered little to no shifts in homesite to indicate residency or commuting, while shifts to multiple homesites towards long-distance migratory elk to indicate migration. We consider both commuting and migration to be migratory coupling. Predictions are summarized in [Figure 1](#).